Jonathan Li & Lanqi Huang
NS 4300
May 13th, 2024

Abstract:

The colloquial saying that stress can make one's hair go gray has been pervasive in culture and the media, but yet, serious scientific validation of this claim has been lacking. Here, through broad and targeted RNA-seq analysis of chemically-induced stressed melanocytes in mice from Zhang et al. (2020), we uncover that stress impacts two spatially distinct melanocyte populations in the hair follicle. Critically, stress depletes both melanocyte populations, leading to the phenotype of gray hairs. Understanding how such cell failure occurs has implications for how stress may potentially affect other organ systems.

Introduction:

Somatic stem cells (SSCs) are undifferentiated cells that can self-renew and yield differentiated cell types that make up tissue. While much is known about how local tissue niches affect SSC behavior, little is known about how systemic physiological states may impact SSC behavior (Zhang et al., 2020).

Hair follicles (HFs) undergo cycles of rest/quiescence, growth, and regression (Chovatiya et al., 2023). Two SSC populations inhabit HFs: hair follicle stem cells (HFSCs) that fuel hair growth and melanocyte stem cells (MeSCs) that provide melanocytes to pigment the growing hair shaft (Sun et al., 2023). During quiescence, MeSCs reside in the dermal papillae of the HF, but upon hair growth, MeSCs differentiate to become transiently amplifying (TA) cells before then splitting its population to either terminally differentiate to mature melanocytes located in the downwards growing hair bulb or translocate back to the upper region of the HF and dedifferentiate into MeSCs, resupplying the stock of MeSCs for future hair cycles (**Figure 1**; Sun et al., 2023). Sympathetic nerves make contact and innervate the HF through the anchorage of the arrector pili muscle to the HF, and upon stress, release noradrenaline, a pertinent neurotransmitter involved in fight-or-flight responses (Zhang et al., 2020).

To investigate how stress impacts MeSCs, Zhang et al. (2020) performed RNA-seq on MeSCs collected from mice that were treated with resiniferatoxin (RTX) or saline. RTX, a chemical analog of capsaicin, initiates pain-induced stress and activates sympathetic nerves (Zhang et al., 2020). Using this dataset, we broadly hypothesize that stress perturbs the transcriptional profile of melanocytes related to activities such as cell migration and death that exhaust/deplete melanocytes from their niche, leading to gray hairs. Understanding the molecular mechanism of stress-induced hair graying could provide insight into treatments for pigmentation disorders but also help us better understand how stress may affect other SSC populations in highly regenerative organs such as intestinal stem cells and muscle stem cells.

Results:

Unbiased, broad hierarchical clustering analysis of the RNA-seq data from Zhang et al. (2020) confirms that, indeed, RTX-induced stressed (RTX) melanocyte samples (n=2) cluster distinctly from control/saline (CT) melanocyte samples (n=2) (**Figure 2**). However, inspection of the heatmap reveals that there is a considerable amount of replicate heterogeneity within the same study arm (CT or RTX) (**Figure 2**), which was not disclosed by Zhang et al. (2020). In an attempt to confirm our retrieved dataset, we attempted to recreate Extended Figure 7e from Zhang et al. (2020). Our recreated heatmap (**Figure 3a**) mainly agrees with the trend demonstrated by the heatmap from Zhang et al. (2020) (**Figure 3b**), in which melanocyte differentiation genes are upregulated upon stress; however, our recreation disagrees with the degree of mRNA expression for certain genes as well as their replicates' homogeneity with each other (**Figure 3a**). One possible explanation of this discrepancy may be due to our potentially different transformations performed on the raw data, but as our recreated heatmap agreed with their trend, we had evidence and more confidence that we were indeed working on the same dataset as described in Zhang et al. (2020), albeit with a different handling. Pathway enrichment analysis with EnrichR on specific clusters from our unbiased heatmap, however, revealed no statistically significant gene ontologies (GOs), and so we decided to pursue other avenues.

Our unbiased, broad volcano plot reveals that there are transcriptional profile changes between the two conditions: RTX and CT (**Figure 4**). To investigate potential biological processes that are perturbed upon stress, we identified genes from the volcano plot that had a statistically significant $\log_2$ 2-times fold change and between 0.5 – 3 on a $-\log_{10}$(p-value) scale between CT and RTX conditions, as colored in red (**Figure 4**). Analysis with EnrichR on this extracted list of significantly upregulated genes upon stress revealed two pertinent biological processes that are enriched in stressed melanocytes: melanogenesis (adjusted p-value = 3.65×10^{-4}) and apoptosis (adjusted p-value = 3.22×10^{-4}) (**Figure 5**).

Stress upregulates melanogenesis genes (*Dct*, *Kit*, *Tyrp1*, and *Ctnnb1*), suggesting that stress promotes committed cell fate determination for MeSCs to fulfill its mature functions (**Figure 6a**). It has been demonstrated that constitutive overexpression of *Ctnnb1*, which encodes for the canonical β-catenin protein of the Wnt pathway, prevents the dedifferentiation of TA cells back into MeSCs (Sun et al., 2023). Compared to CT melanocytes, stressed melanocytes have upregulated *Ctnnb1* expression, which may play a role in preventing a greater proportion of the TA cell population from translocating and dedifferentiating back to renew the MeSC population in the upper region of the HF (**Figure 6b**). Additionally, stressed and committed melanocytes upregulate the expression of apoptosis genes (*Lmna*, *Ctsd*, *Actb*, *Actg1*, and *Ctsb*), depleting the mature melanocyte population in the hair bulb (**Figure 7**).

Discussion:

Our RNA-seq analysis suggests that stress inhibits the renewal of the MeSC population through the upregulation of terminal differentiation (preventing dedifferentiation) as well as the depletion of the mature melanocyte population through the promotion of apoptosis, which upon further hair cycles that repeatedly challenge the MeSCs' ability to renew its own population after each transient differentiation program may ultimately lead to hair graying (**Figure 8**). This example illustrates how stress can drive the loss of SSCs and raises the point on how there is more to consider when examining stem cell behavior beyond its local tissue niches/environments that surround it: that the maintenance of SSCs may also be influenced by the overall physiological state of the organism — an interesting avenue for future investigation for other SSCs found in other organs beyond the skin and hair.

However, our analyses and the dataset from Zhang et al. (2020) has limitations to note. Critically, the RNA-seq data from Zhang et al. (2020) severely lacks statistical power, with only two samples (n=2) for each study condition (CT and RTX). This results in significant variability observed within the same study condition, which can be observed in our broad unbiased heatmap (**Figure 1**), and also the decreased ability to make statistically significant claims about the actual perturbed biological processes in stressed melanocytes compared to control. Additionally, the RNA-seq dataset was from mice, but the pathway enrichment analysis with EnrichR was performed using the KEGG 2021 Human Pathways database. This contradiction of organisms (mouse and human) between data collection and analysis also limits the utility of our analyses, as mice and human biology differ.

Nevertheless, our analyses do attempt to bridge together current knowledge of melanocyte dynamics in the hair follicle from Sun et al. (2023), specifically the aspect of translocation and dedifferentiation for MeSC renewal, with the RNA-seq dataset from Zhang et al. (2020) that had previously considered MeSCs to be a spatially permanent cell population in the hair follicle. We thus attempt to update the field's understanding of how stress impacts the melanocyte stem cell system.

Methods:

Raw data on the number of transcript reads per gene and gene length was accessed from the Gene Expression Omnibus database via the National Center for Biotechnology Information (NCBI) with the accession code: GSE131566. Data was normalized using the transcripts per million (TPM) formula. Heatmaps were generated with Pearson correlation as the distance metric and the average linkage method. P-values for volcano plots were computed with a two-tailed, equal variance Student's t-test and corrected for multiple tests by using the Bejamini-Hochberg procedure and setting the false discovery rate (FDR) to 5%. Significant upregulated genes upon stress were defined to be above a $\log_2$ 2-times fold change for the ratio of mRNA abundance between control and RTX conditions and between 0.5 - 3 on a $-\log_{10}$ (p-value) scale. EnrichR and its KEGG 2021 Human Pathways Database were used for pathway enrichment analysis. Boxplots were generated to display differential gene expressions within relevant gene ontology categories with TPM data subjected to $\log_{10}$ scaling for more succinct visualization. Boxes represent the interquartile range (IQR, which is the 25th to 75th percentile). The bottom and top whiskers display the range of 1.5xIQR below and above the 25th and 75th percentiles, respectively.

References

Chovatiya, G., Li, K.N., Li, J. *et al.* Alk1 acts in non-endothelial VE-cadherin+ perineurial cells to maintain nerve branching during hair homeostasis. *Nature Communications* 14, 5623 (2023). https://doi.org/10.1038/s41467-023-40761-5

Sun, Q., Lee, W., Hu, H. et al. Dedifferentiation maintains melanocyte stem cells in a dynamic niche. Nature 616, 774–782 (2023). https://doi.org/10.1038/s41586-023-05960-6

Zhang, B., Ma, S., Rachmin, I. et al. Hyperactivation of sympathetic nerves drives depletion of melanocyte stem cells. Nature 577, 676–681 (2020). https://doi.org/10.1038/s41586-020-1935-3

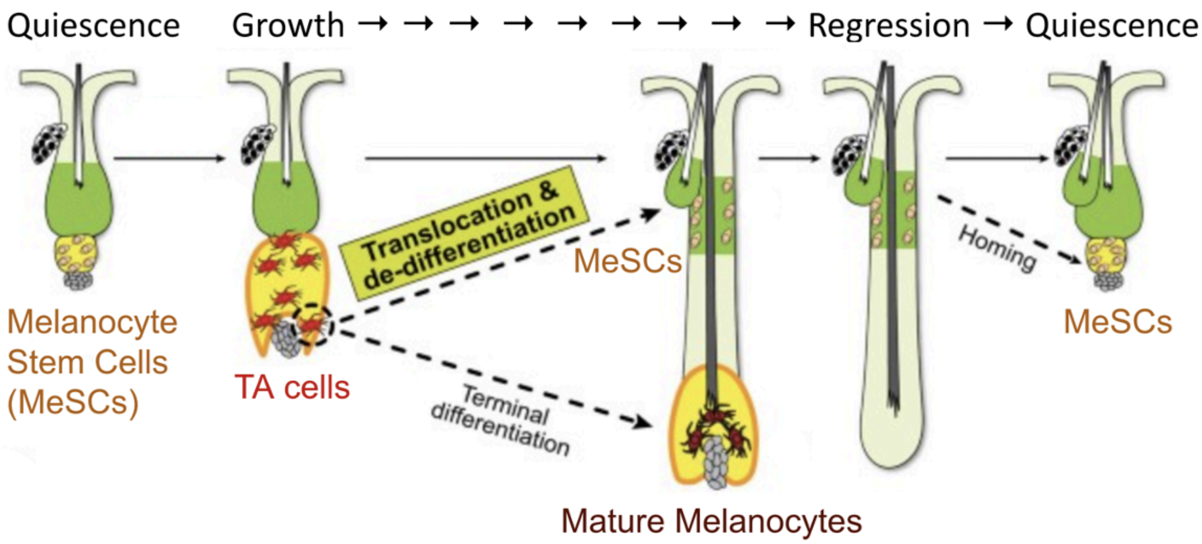

Figure 1. The Hair Cycle and the Melanocytes' Dynamic Niche
Cartoon displaying the differentiation program and niches of melanocyte stem cells and their progeny during the hair cycle as refined by Sun et al., *Nature*, 2023.

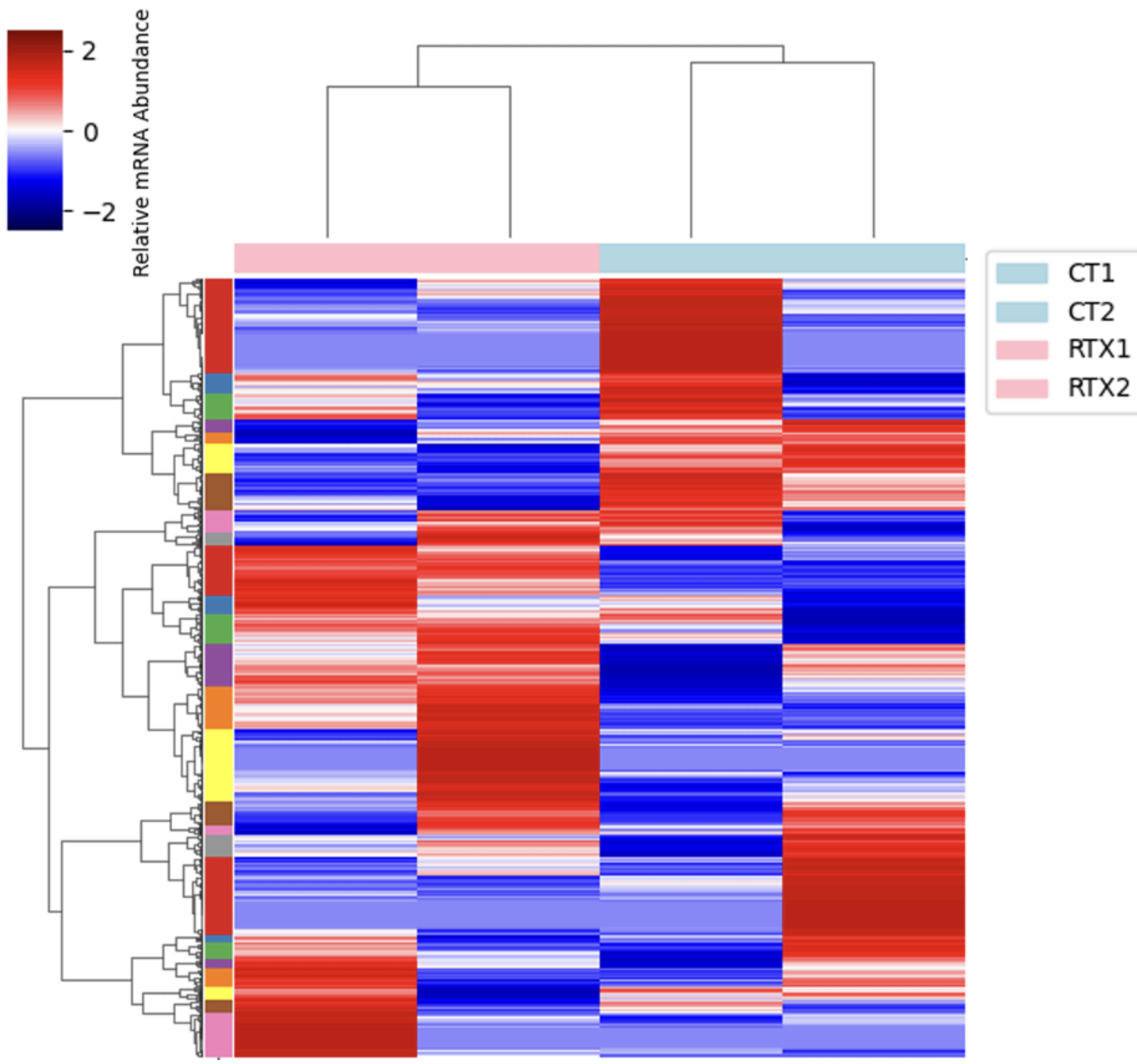

Figure 2. Heat map of relative mRNA abundances amongst control and RTX-treated melanocyte samples from mice

Broad, unbiased heat map of the relative mRNA abundances amongst control and RTX-treated melanocyte samples from mice. mRNA abundances were TPM normalized. Pearson correlation and average linkage with the distance metric and linkage method used, respectively, for the hierarchical clustering of the samples.

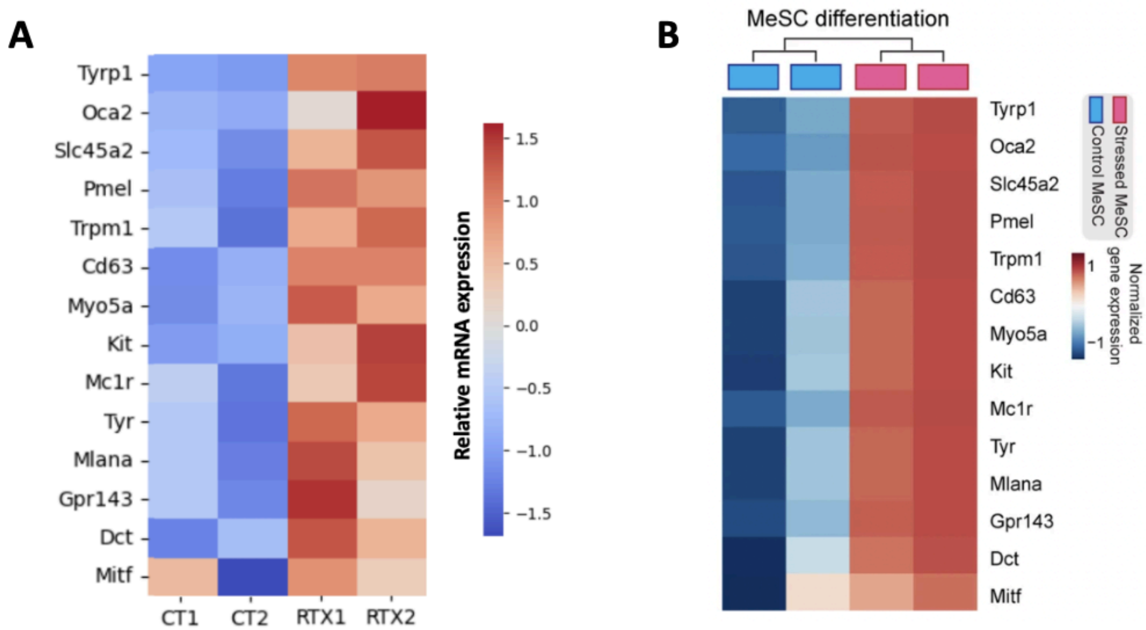

Figure 3. Confirmation of the dataset from Zhang et al. (2020)

A) Recreated heat map of relative melanocyte stem cell differentiation gene expressions for CT and RTX samples, using the same genes as Zhang et al. (2020) in its Extended Figure 7e. mRNA abundances were TPM normalized. Pearson correlation and average linkage with the distance metric and linkage method used, respectively, for the hierarchical clustering of the samples.

B) Extended Figure 7e. from Zhang et al. (2020)

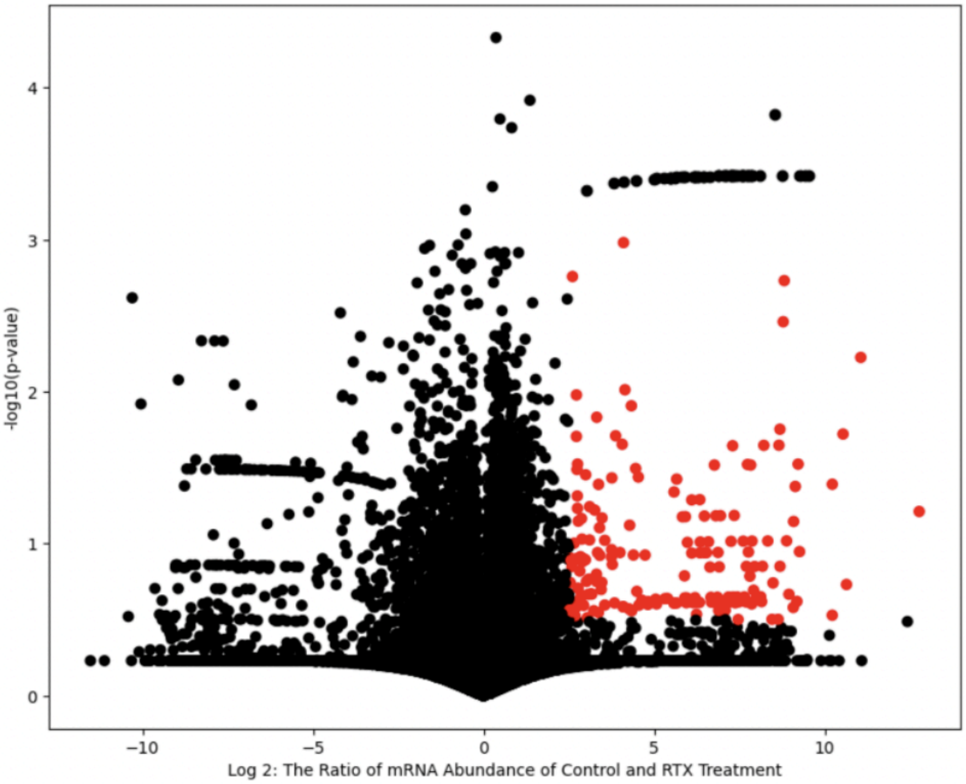

Figure 4. Volcano plot of the Control vs. RTX treated melanocyte sample mRNA abundances with significantly upregulated genes upon stress highlighted in red

Volcano plot of the log2 fold change of mRNA abundance in the RTX treated melanocyte samples to the control melanocyte samples against its $-\log_{10}$(p-value), where the p-value was computed with a two-tailed, equal variance Student's t-test. The p-values were corrected for multiple tests, using the Benjamini-Hochberg procedure and setting the false discovery rate to 5%. Significantly upregulated genes upon stress, as defined in methods, are plotted in red.

GO Term	Genes from significantly upregulated genes upon stress	Adjusted P-value
Melanogenesis	Ednrb, Dct, Kit, Tyrp1, Ctnnb1, Calm2	3.65E-04
Apoptosis	Jun, Lmna, Fos, Ctsd, Actb, Actg1, Ctsb	3.22E-04

Figure 5. Table of gene ontology terms and their genes that are part of the significantly upregulated genes upon stress list as well as their adjusted p-value from EnrichR
Table of selected biologically pertinent/significantly enriched pathways (melanogenesis and apoptosis) and its members found in significantly upregulated genes upon stress in melanocytes as well as its adjusted p-value from the pathway enrichment analysis using EnrichR and the KEGG 2021 Human Pathways Database.

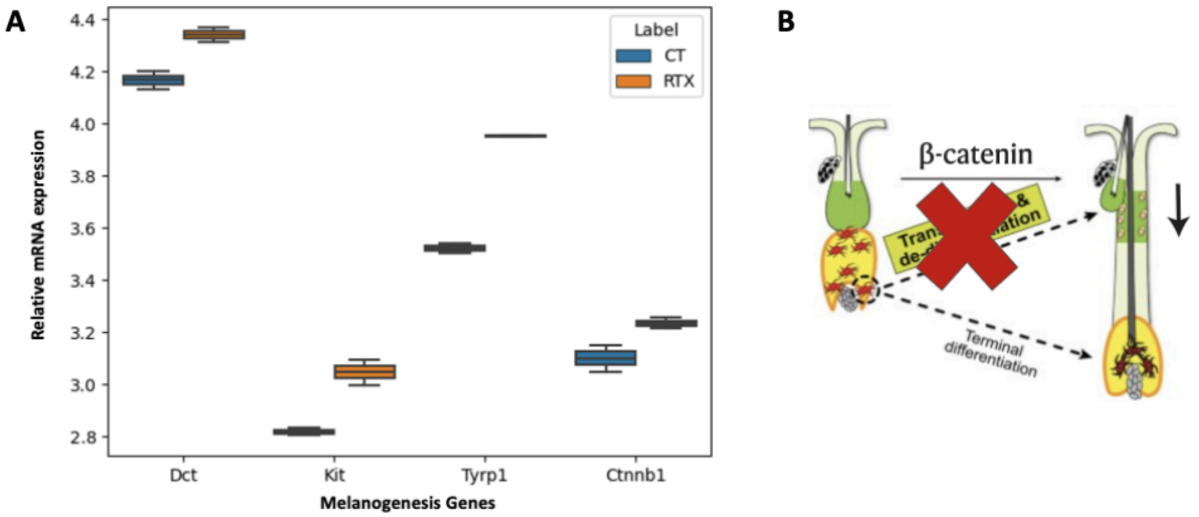

Figure 6. The relative mRNA expression of melanogenesis genes between control and RTX treated melanocytes

A) Boxplots displaying the median relative mRNA abundance of Dct, Kit, Tyrp1, and Ctnnb1 for control and RTX treated melanocyte samples from mice. Transcript data was TPM normalized. Boxes represent the interquartile range (IQR, which is the 25^{th} to 75^{th} percentile). The bottom and top whiskers of the boxplot display the range 1.5xIQR below and above the the 25^{th} and 75^{th} percentiles, respectively.

B) Cartoon summarizing the stress-induced greater expression of Ctnnb1 prevents translocation and dedifferentiation to renew the MeSC population

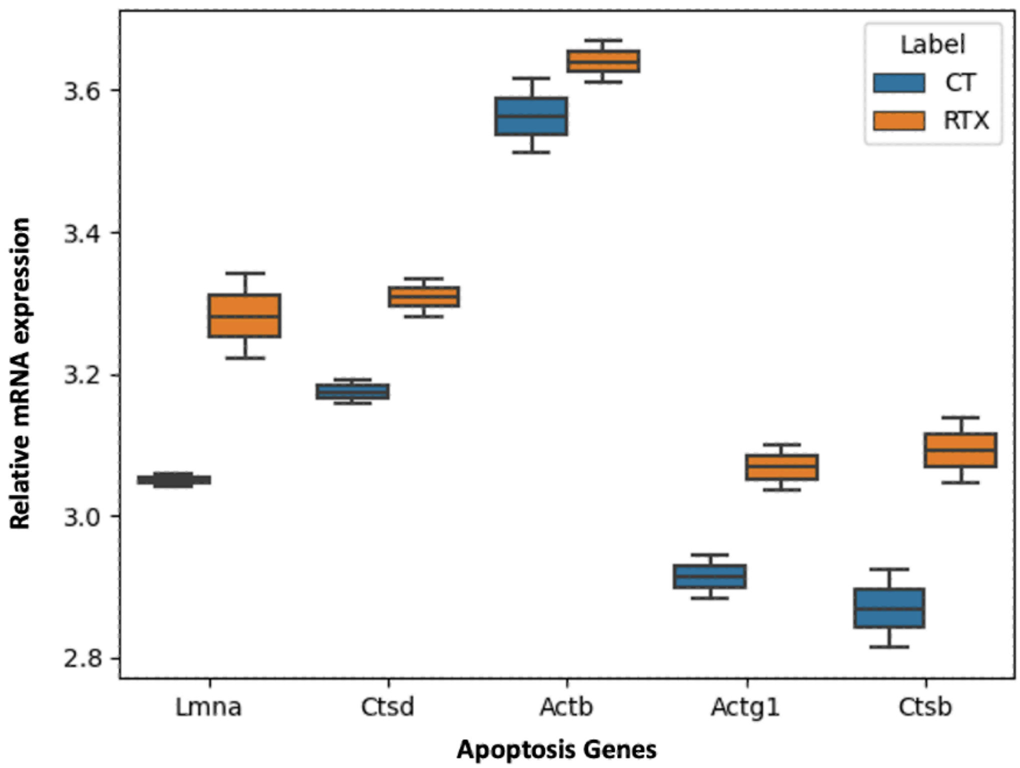

Figure 7. The relative mRNA expression of apoptosis genes between control and RTX treated melanocytes
Boxplots displaying the median relative mRNA abundance of Lmna, Ctsd, Actb, Actg1, and Ctsb for control and RTX treated melanocyte samples from mice. Transcript data was TPM normalized. Boxes represent the interquartile range (IQR, which is the 25^{th} to 75^{th} percentile). The bottom and top whiskers of the boxplot display the range 1.5xIQR below and above the the 25^{th} and 75^{th} percentiles, respectively.

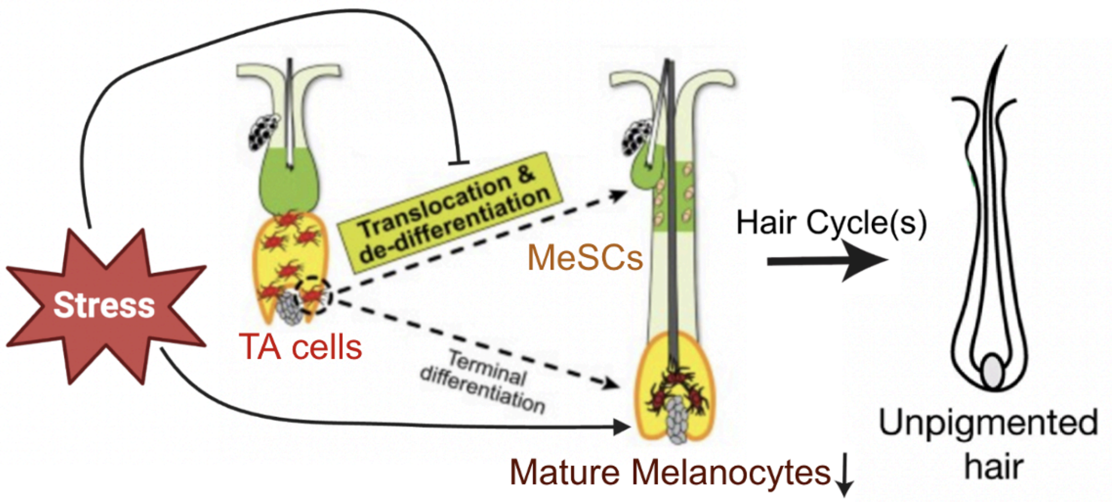

Figure 8. Stress impacts two spatially distinct melanocyte populations in the hair follicle
Cartoon summarizing how stress inhibits the renewal of the MeSC population in the upper region of the hair follicle and depletes the mature melanocyte population in the hair bulb, using current knowledge of melanocyte dynamics in the hair follicle from Sun et al., *Nature*, 2023.